

# **Automobile Analysis Dashboard Report**

## **Introduction:**

This project focuses on analysing automobile sales data using **Power BI** to gain insights into brand performance, regional sales trends, and overall profitability.

The dataset includes details such as *Make, Model, Vehicle Type, Country, Sale Price, Profit, Cost, and Invoice Date* from multiple years.

The main objective of this project is to clean and transform the dataset using **Power Query**, create calculated measures using **DAX**, and develop interactive **Power BI dashboards** that visually represent key business metrics to assist in decision-making.

## **Methodology:**

The project was developed in Power BI, following these steps:

1. **Data Loading:** Imported the automobile dataset into Power BI Desktop.
2. **Data Cleaning & Transformation:**
  - Used **Power Query Editor** to filter data for specific brands (like Jaguar), remove unnecessary columns, and replace inconsistent values.
  - Created new calculated columns such as **TotalCost = CostPrice + DeliveryCharge + SpareParts + LaborCost**.
  - Split InvoiceDate into InvoiceYear and InvoiceMonth for time-based analysis.

- Grouped data by Make and Country name to summarize total sales.
- 3. Data Modelling:**
- Built relationships between table
  - Created DAX measures for Total Sales, YTD Sales, Average Sales per month, Average sales per customer, Profit, Profit Margin Percent etc...
- 4. Visualization:** Built multiple charts like bar, line, pie, card visuals to represent insights.
- 5. Dashboard Creation:** Combined visuals into one interactive dashboard for analysis.

### **Requirement Analysis:**

- 1. Software Requirements:**
- Power BI Desktop
  - Microsoft Excel for Data
- 2. Data Requirements:**
- Fields used are Make, Model, VehicleType, CountryName, SalePrice, Profit, CostPrice, DeliveryCharge, SpareParts, LaborCost, InvoiceDate
- 3. Analytical Requirements:**
- Analyse total and year-wise sales performance.
  - Compare actual vs. budgeted sales across years.
  - Evaluate regional and vehicle-type-based trends.
  - Measure profit margin and sales growth percentage
  - Identify top-selling brands and products.

### **Other Parameters:**

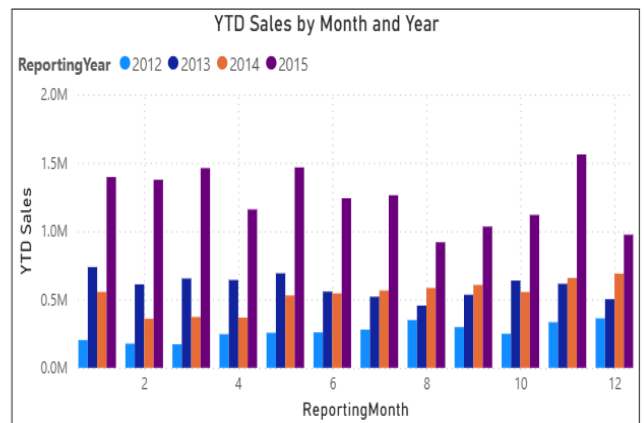
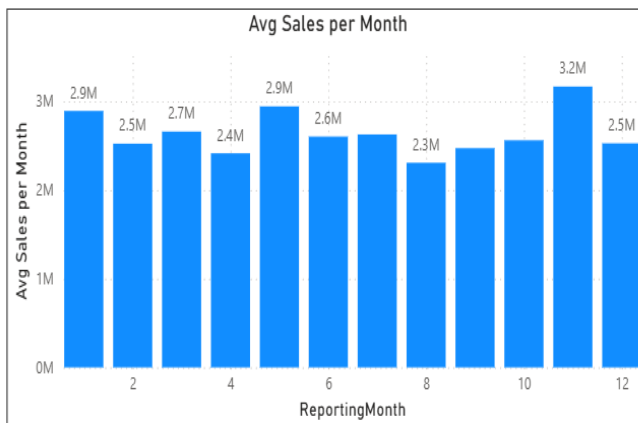
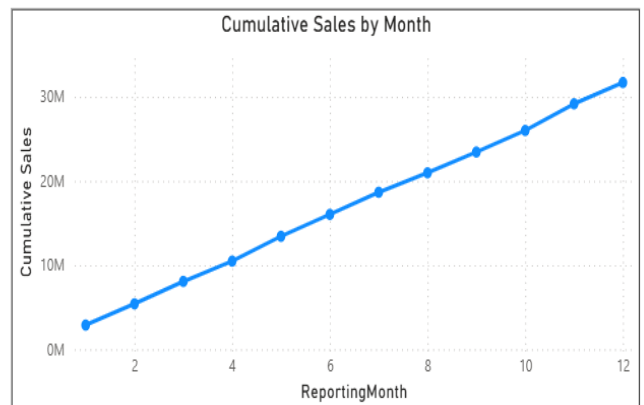
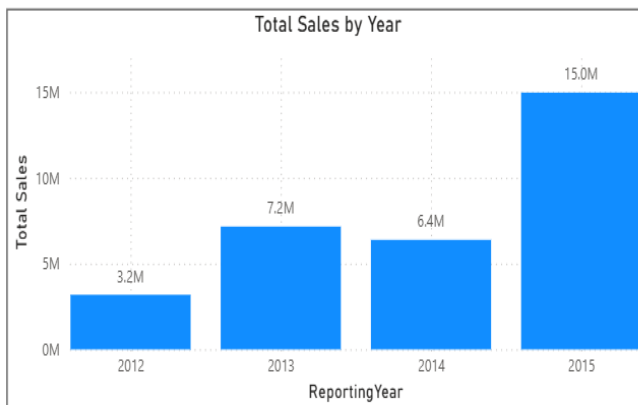
In this project, several data transformation and analytical steps were performed using Power Query and DAX to prepare the automobile dataset for effective visualization and insight generation.

The following operations were done as a part of data transformation:

- 1. Data Cleaning & Filtering:** Columns such as labour Cost and Client name were removed from the table by referencing it from the source table. Data was filtered for specific automobile brands such as Jaguar by referencing the source data. All the duplicates of the Invoice date and Make were removed by referencing the source data.
- 2. Date based Transformation:** The invoice date was split into Invoice Year and Invoice Month to enable Year wise and Month wise trend analysis.
- 3. Grouping and Aggregation:** The data was grouped based on Make and Country name to get the Sale price for each group.
- 4. Duplicate Removal:** Duplicate records were identified and removed based on InvoiceDate and Make to ensure accurate sales and profit totals.
- 5. Budget Table Creation:** A new Budget Table was created using DAX for years 2012–2015 with incremental growth of 5%, 10% and 15% respectively. This enabled comparison between actual vs budgeted sales and identification of sales variances.
- 6. KPI and Performance measures:** Several DAX measures were added to track business metrics such as total sales, Total Profit, Profit Margin %, YTD Sales, Sales Variance %, Sales Growth % (YoY)

**7. Visualisation Preparation:** Multiple Power BI visuals such as bar charts, line charts, maps, tree maps, and histograms were designed with consistent colour schemes, titles, and data labels for better readability and interpretation. Interactive slicers were added for dynamic filtering by brand, country, and year.

### Visualisation:



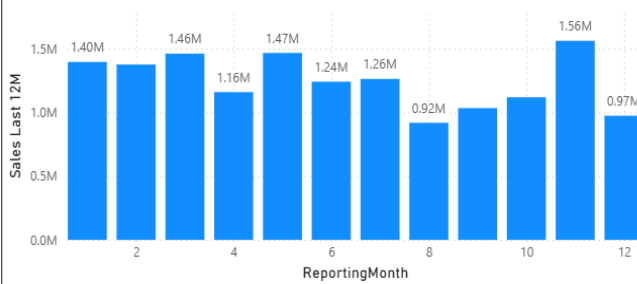
0.89

Sales Growth %

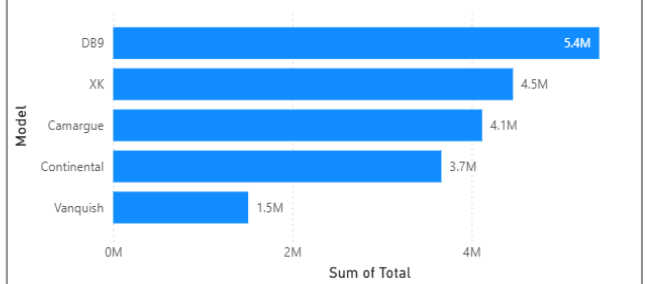
3M

Highest Sales for a Month

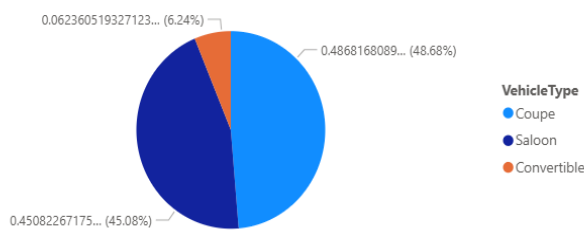
Sales of last 12 Months



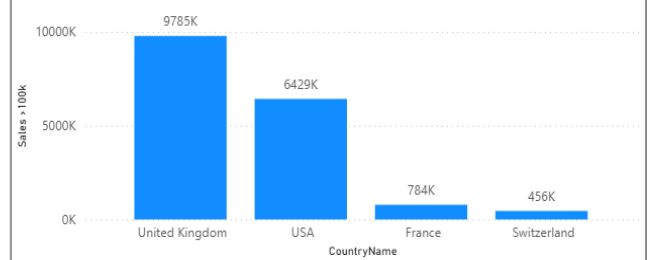
Top 5 sales by Model



Sales Contribution % by VehicleType



Sales &gt; 100k by Country



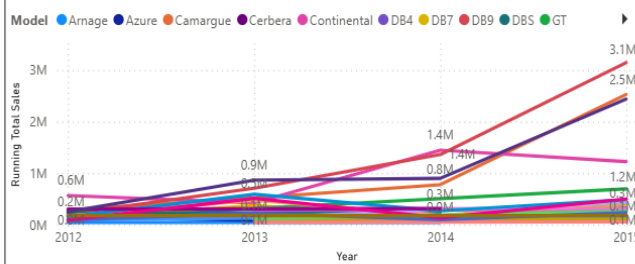
31

Distinct Customers

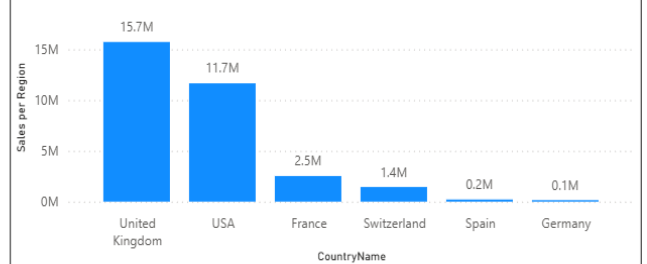
6M

Sales Jaguar

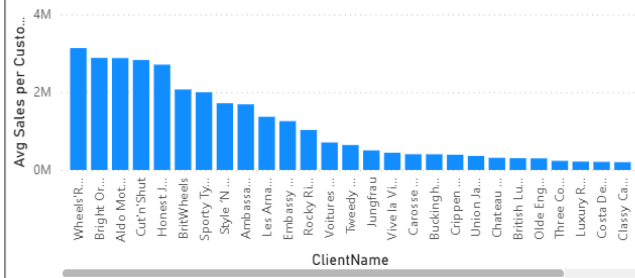
Running Total Sales by Product



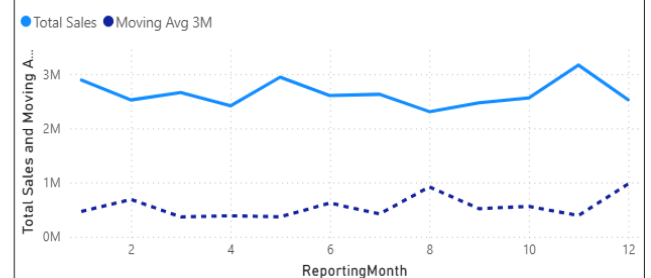
Sales per Region



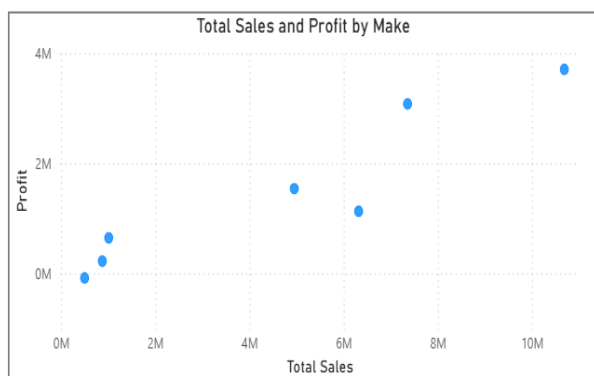
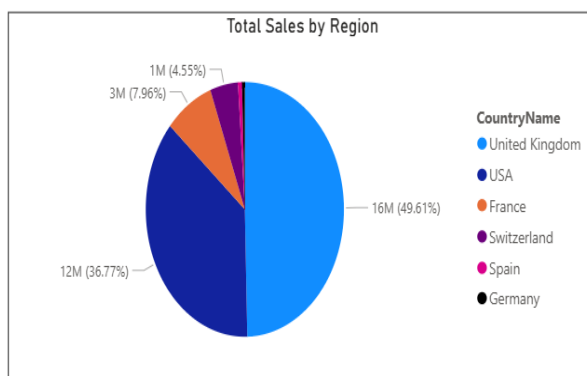
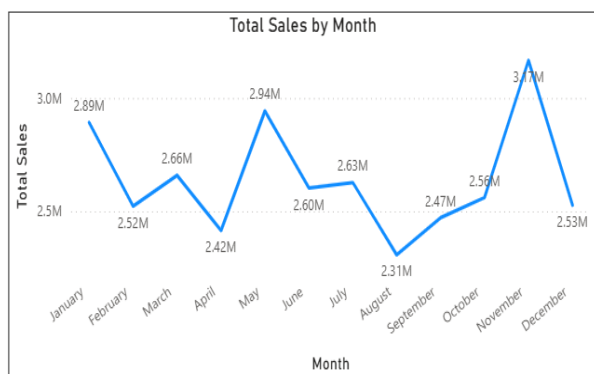
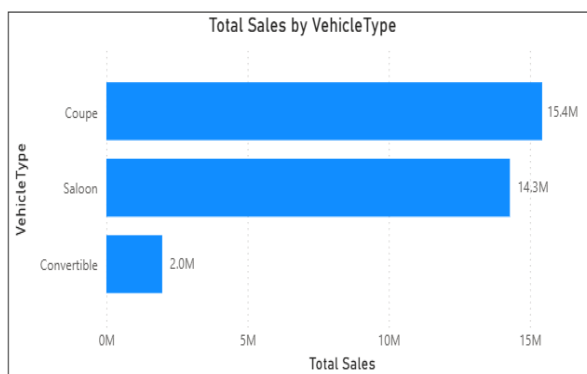
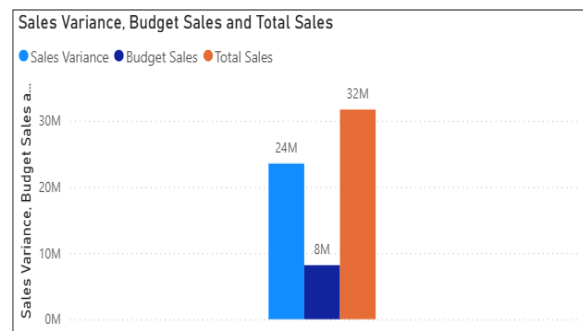
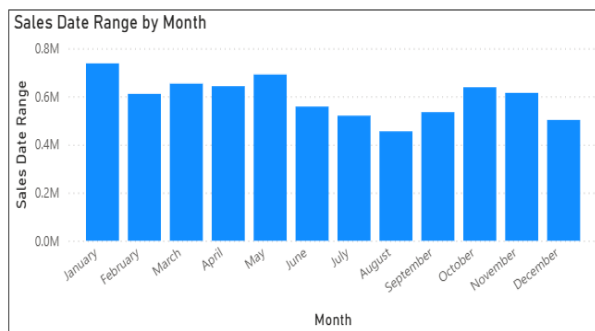
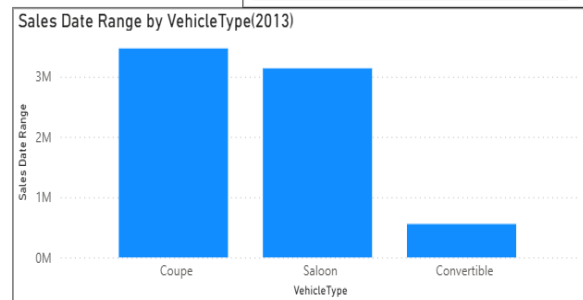
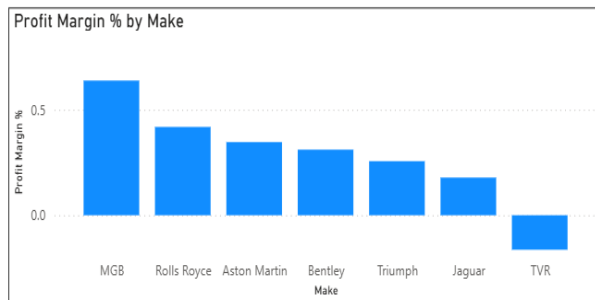
Avg Sales per Customer

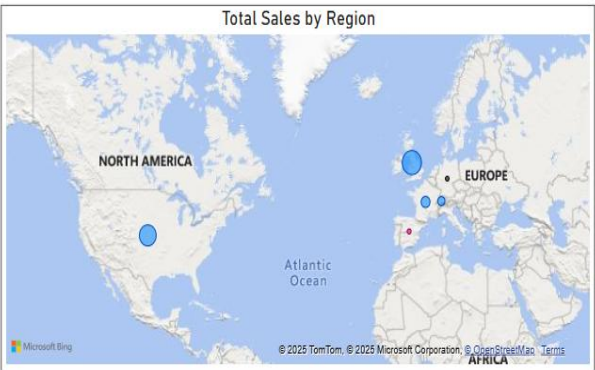
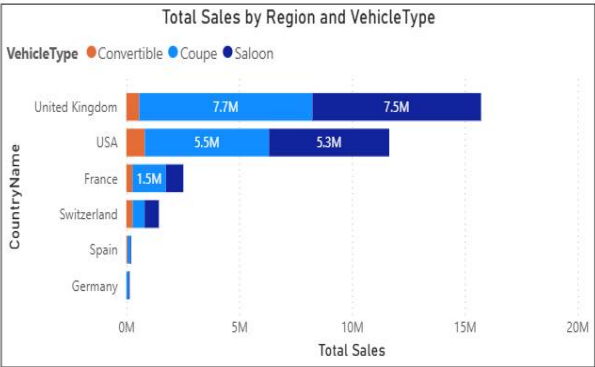
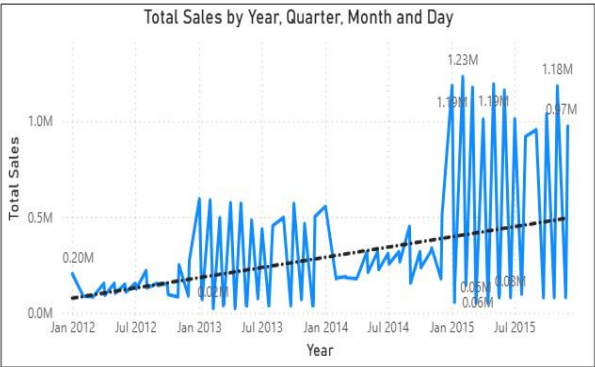


Total Sales and Moving Avg 3M by ReportingMonth

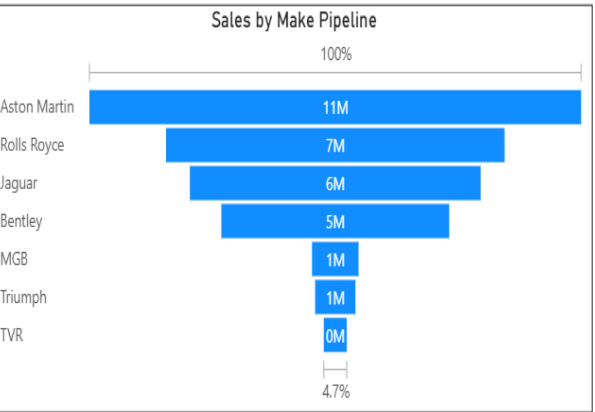
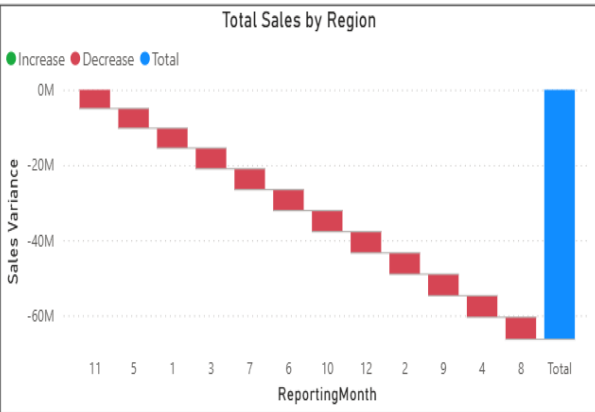
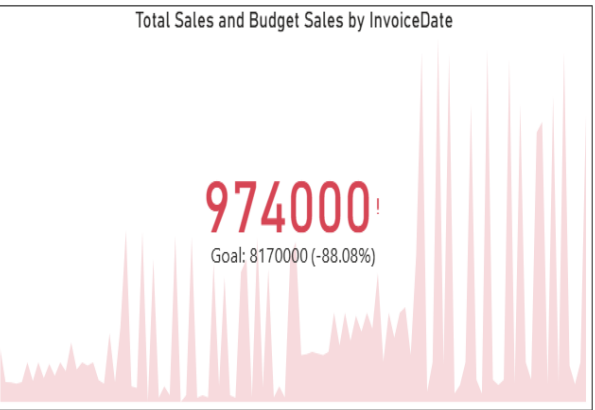
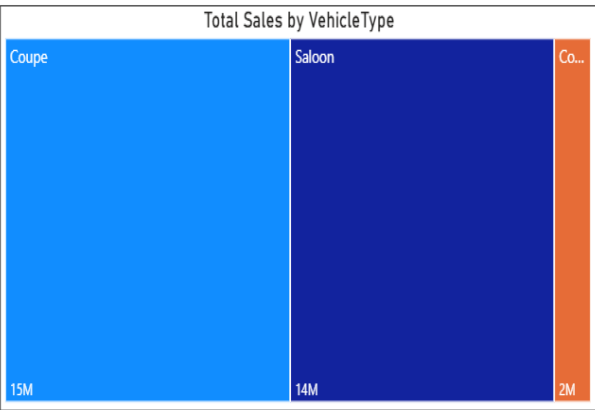


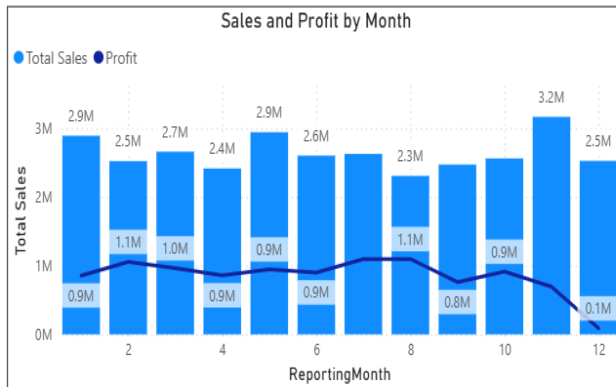
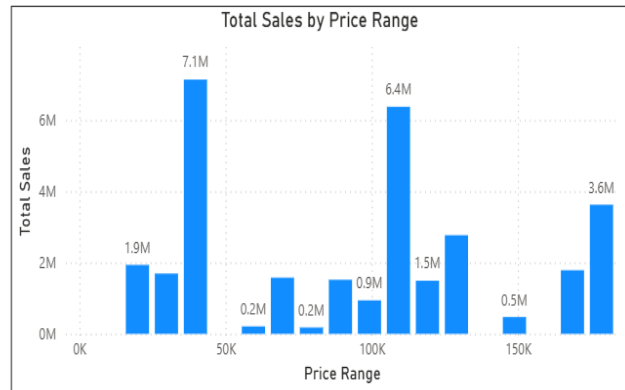
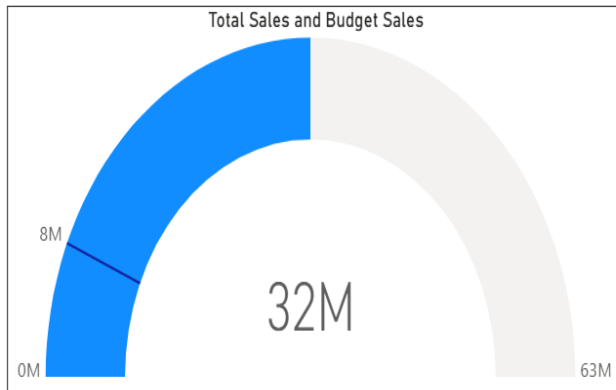
| Model        | Sales Rank | Total Sales     |
|--------------|------------|-----------------|
| Amage        | 25         | 90750           |
| Azure        | 17         | 489500          |
| Camargue     | 3          | 4116900         |
| <b>Total</b> | <b>1</b>   | <b>31697940</b> |





| CountryName    | Convertible | Coupe    | Saloon   | Total    |
|----------------|-------------|----------|----------|----------|
| France         | 267570      | 1484890  | 772050   | 2524510  |
| Germany        |             | 103500   | 42250    | 145750   |
| Spain          | 46750       | 121500   | 39500    | 207750   |
| Switzerland    | 273500      | 536170   | 631300   | 1440970  |
| United Kingdom | 573500      | 7674750  | 7476750  | 15725000 |
| USA            | 815380      | 5510280  | 5328300  | 11653960 |
| Total          | 1976700     | 15431090 | 14290150 | 31697940 |





**32M**

Total Sales

**VehicleType**

|                          |             |
|--------------------------|-------------|
| <input type="checkbox"/> | Convertible |
| <input type="checkbox"/> | Coupe       |
| <input type="checkbox"/> | Saloon      |

| Make                | 2012           | 2013           | 2014           | 2015            | Total           |
|---------------------|----------------|----------------|----------------|-----------------|-----------------|
| <b>Aston Martin</b> | <b>1152000</b> | <b>2355660</b> | <b>2262440</b> | <b>4915940</b>  | <b>10686040</b> |
| DB4                 | 68500          | 322500         | 71250          | 330750          | 793000          |
| DB7                 | 125000         | 375740         | 127750         | 394990          | 1023480         |
| DB9                 | 205000         | 710070         | 1361940        | 3146850         | 5423860         |
| DBS                 | 230000         |                | 235500         |                 | 465500          |
| Rapide              | 95000          | 130000         | 97750          | 132750          | 455500          |
| Vanquish            | 155000         | 591500         | 270500         | 489750          | 1506750         |
| Vantage             | 95000          | 225850         | 97750          | 239600          | 658200          |
| Zagato              | 178500         |                |                | 181250          | 359750          |
| <b>Bentley</b>      | <b>919500</b>  | <b>702500</b>  | <b>1811500</b> | <b>1517750</b>  | <b>4951250</b>  |
| Arnage              | 44000          |                | 46750          |                 | 90750           |
| Azure               | 154000         | 85250          | 159500         | 90750           | 489500          |
| Continental         | 567500         | 423750         | 1445750        | 1225250         | 3662250         |
| Turbo R             | 154000         | 193500         | 159500         | 201750          | 708750          |
| <b>Jaguar</b>       | <b>647500</b>  | <b>1380000</b> | <b>1352000</b> | <b>2939500</b>  | <b>6319000</b>  |
| <b>MGB</b>          |                | <b>315000</b>  |                | <b>696000</b>   | <b>1011000</b>  |
| <b>Rolls Royce</b>  | <b>466500</b>  | <b>1947300</b> | <b>960500</b>  | <b>3982600</b>  | <b>7356900</b>  |
| <b>Triumph</b>      |                | <b>286500</b>  |                | <b>588500</b>   | <b>875000</b>   |
| <b>TVR</b>          |                | <b>171750</b>  |                | <b>327000</b>   | <b>498750</b>   |
| <b>Total</b>        | <b>3185500</b> | <b>7158710</b> | <b>6386440</b> | <b>14967290</b> | <b>31697940</b> |



## Insights from Charts and Dashboard

### **1. Sales Insights**

- Total sales show a **steady year-on-year growth** from 2012 to 2015, increasing from **3.2 M** in 2012 to nearly **15 M** in 2015.
- The largest rise in sales occurred between 2014 and 2015, with an increase of 8.6M.
- Cumulative and YTD sales visuals confirm consistent month-over-month growth, with the highest sales peaks toward the end of each year.
- Overall sales growth remained positive which is 0.89 %.

### **2. Category Insights (Vehicle Type and Model):**

- Among all vehicle types, Coupe contributes to the highest shares of 48.68% followed by Saloons with a share of 45.08 % and Convertibles with shares of 6.24 %.
- The top five performing models are DB9, XK, Camargue, Continental, and Vanquish, with DB9 leading at 5.4 M in total sales.
- The majority of vehicles are priced between 50 K – 150 K, indicating a balanced mid-range market.

### **3. Regional Insights:**

- United Kingdom records the highest sales value which is 15.7M, followed by the USA 11.7 M.
- France and Switzerland perform moderately, while Germany and Spain contribute comparatively less.
- The UK and USA together make up over 80 % of total sales.

- Convertibles show relatively higher demand in Switzerland, while Coupes and Saloons dominate in the UK and USA.

#### **4. Profit and Budget Insights:**

- Rolls Royce and Aston Martin exhibit the highest profit margins, followed by Bentley and Jaguar.
- TVR and Triumph yield lower margins, indicating limited presence in the luxury segment.
- Profit trends follow sales closely — months with higher sales also record proportionally higher profits
- Actual sales consistently exceeded budget targets, particularly in the third and fourth quarters.

#### **5. Customer Insights:**

- Analysis across 31 unique customers shows varied purchasing behavior.
- A few high-value clients such as *Luxury Rides* and *Ambassador Motors* contribute disproportionately higher sales, indicating repeat luxury buyers.
- Average sales per customer reveal that premium clients maintain higher-than-average spending, strengthening overall profitability.

#### **6. Performance and Trend insights:**

- The 3-month moving average remains close to actual sales which shows stable sales pattern.
- Sales and profit trends both display progressive upward slopes, suggesting a well-balanced pricing and marketing strategy.
- Total cumulative sales across all years reached approximately 32M.

This Automobile Dashboard clearly demonstrates steady growth, strong regional performance, and healthy profitability across four years.

Luxury brands like Aston Martin, Rolls Royce, and Jaguar dominate margins, while Coupes and Saloons lead in sales volume.

Overall, the company consistently met and exceeded its budget expectations.

### **Conclusion:**

The Automobile Dashboard Analysis Project effectively showcases how Power BI can convert raw automobile sales data into valuable insights.

By using Power Query for data cleaning, DAX for calculations, and interactive visualizations, the project provides a clear view of sales, profit, and market performance.

The analysis revealed steady year-on-year sales growth, with Coupes and Saloons as top-selling vehicle types and Aston Martin, Rolls Royce, and Jaguar leading in profitability.

The UK and USA emerged as key markets, and actual sales consistently exceeded budget targets, reflecting strong market demand and effective sales strategies.

Overall, the project demonstrates the power of data-driven decision-making in identifying trends, improving strategy, and driving business growth in the automobile sector.